

Testing: Consistency of Newton's Universal Law of Gravitation and Kinematic Equations by The Acceleration due to Gravity.

Std. Enes ÇİÇEK

Şehit Öğretmen Necmeddin Kuyucu Anatolian High School
Gebze, Kocaeli, Türkiye

Abstract—In this article, the consistency of the Universal Law of Gravitation with Kinematic Equations, the effect of air resistance on kinematic equations, the temporal error margin of human hand reflexes, and the behavior of different objects in kinematic equations were determined through free-fall tests in controlled conditions. The consistency of the gravitational acceleration equation with kinematic equations was measured at 83.20%, with the effect of air resistance on this average being 4.29%, and the human error margin depending on altitude was measured at 16.80%.

Index Terms—Gravity, Acceleration, Force, Mass, Time

I. INTRODUCTION

The gravity acceleration test involves the quantitative measurement of “acceleration due to gravity”, a vector quantity in Newtonian physics, using a simple experimental setup on different materials. During the experiment, the relationship between time, velocity, and acceleration will be established by dropping a pencil sharpener (Rectangle prism), an eraser (Approximately rectangular prism), and a pencil (Cylinder) from different heights under identical conditions. The primary objective of this experiment is to investigate whether the gravitational acceleration value g , derived theoretically from Newton's Law of Universal Gravitation, is consistent with the value obtained experimentally through kinematic equations [11]. The degree of consistency between these two values will be evaluated through percent error analysis. Gravity acceleration is a vector quantity that was first defined conceptually by Giuseppe Moletti at 16th century [1] and mathematically by Isaac Newton towards the end of the 17th century. According to the universal law of gravitation, in an environment where friction is neglected, every object is attracted toward the center of mass of the celestial body it is on with equal acceleration. The gravitational force is inversely proportional to the square of the object's distance (d) from the planet and directly proportional to the planet's mass (M), object's mass (m) and the universal gravitational constant (G).

$$F = mg = G \frac{Mm}{d^2} \quad (1)$$

$$g = G \frac{M}{d^2} (M \gg m) \quad (2)$$

The formula used in the equation (2) is a simplified version of the mathematical model of the “Universal Law of Gravitation” developed by Isaac Newton [2], applied to negligible masses. This experiment was conducted at a distance of $d \approx 6.367 \times 10^6 m$ from the Earth's center of mass. The Universal Constant of Gravitation (G) is $6.674 \times 10^{-11} m^3/kg s^2$ and the mass of Earth is $5.972 \times 10^{24} kg$ [3], [4]. When these values were added to the equation, the gravitational acceleration of the experimental environment, $g \approx -9.827 m/s^2$, was determined¹. This will be the value expected to be obtained as a result of the experiments conducted. Air resistance was also taken into account in the calculations conducted in the experimental environment. To calculate air resistance, temperature and ambient atmospheric pressure were measured. The temperature of the experimental environment (T) was measured with an infrared thermometer at 300 K, and the atmospheric pressure (P_0) in the experimental environment was measured with a barometer at $1.014 \times 10^5 Pa$. For the average molar mass (M_{air}) of the gasses in the environment, which is the average molar mass in dry air. $M_{air} = 0.029 kg/mol$ it is based on.

The data obtained in this experiment will measure the mathematical and practical consistency of the acceleration due to gravity. The drag coefficient (C_d), which is determined by the geometric shapes of objects, has not been calculated based on the short-distance throw of objects and their inability to reach the terminal velocity. For the drag coefficient, values given in the literature for different geometric shapes will be used. The formula for net force acting on objects is as follows:

$$F_{net} = ma = mg - F_{air} \quad (3)$$

$$m \frac{dv}{dt} = mg - \frac{1}{2} C_d \rho v^2 \quad (4)$$

$$k = \frac{C_d \rho A}{2m} \quad (5)$$

$$\frac{dv}{dt} = g - kv^2 \quad (6)$$

¹Neglecting any other gravitational forces.

To find terminal velocity:

$$g = -kv^2 \longrightarrow v_t = \sqrt{\frac{g}{k}} \quad (7)$$

$$\frac{dv}{dt} = g(1 - \frac{v^2}{v_t^2}) \quad (8)$$

$$\int \frac{1}{1 - \frac{v^2}{v_t^2}} dv = \int g dt \quad (9)$$

$$v(t) = v_t \tanh(\frac{gt}{v_t}) \quad (10)$$

In equation (3) [5]–[7], A , which has not been defined yet, is the cross-sectional area of the impact section of the object and v is the instantaneous velocity of the object. Here, since the $-y$ direction is considered **positive** and the $+y$ direction is considered **negative**, the g value must be negative. If we adapt the equation (3) to the kinematic equations, we can obtain h in the following step:

$$h(t) = \int v(t) dt = \frac{v_t^2}{g} \ln(\cosh(\frac{gt}{v_t})) \quad (11)$$

Since the objects are released without initial velocity, the v_0 value was not included.

The function $h(t)$ is the height at which the objects are left. The variable t is the time that objects spend until they hit the ground.

Of course, here it is assumed that only the air resistance and gravity accelerations act on the object and all others are neglected.

II. RELATED WORKS

Physicist and Mathematician Isaac Newton published a book called *Mathematical Principles of Natural Philosophy* in 1687, where he announced the Law of Universal Gravitation to the world. This article is based on this book.

Similar to the experiment conducted in this article, the acceleration due to gravity was calculated four times in an experimental setup developed with an Arduino UNO board in 2020 [8].

Similarly, it is observed that the acceleration due to gravity is proven in an experiment conducted with a pendulum motion in 1965 [9].

It was observed that a similar study was conducted in 1965 using a special gravity apparatus in NIST [10].

There are many articles on this subject in the literature. The difference of this article is that these measurements were performed manually using amateur equipment and the air resistance value was taken into account. Additionally, the margin of error was also considered.

TABLE I
VARIABLES

Independent Variables	$[h], [A], [m], [C_d],$
Dependent variables	$[t], [g]$
Controlled Variables	$[T], [P_0], [\rho], [v_t]$

III. METHOD

A. Dependent, Independent and Controlled Variables.

v is an intermediate variable derived from t and g .

The reason why the t value is the dependent variable is that it occurs as a result of a measurement and is included in the equation to give the g value. In other words, the g value is actually also dependent on t .

B. The Values of The Variables

TABLE II
THE SURFACE AREAS OF TESTING OBJECTS

Objects	A (m^2)
Pencil Sharpener	1.500×10^{-4}
Eraser's Largest Surface Area	8.000×10^{-4}
Eraser's Medium-sized Surface Area	4.000×10^{-4}
Eraser's Smallest Surface Area	2.000×10^{-4}
Pencil's Largest Surface Area	1.400×10^{-3}
Pencil's Smallest Surface Area	0.785×10^{-4}

The area values of the surfaces used by the tested objects are specified in Table II. The height variable from which each object is dropped will also be considered in two different cases. In the first case, each object will be dropped from a height of $h = 0.3 m$, while in the second case, it will be dropped from a height of $h = 1 m$. The measured face of the sharpener is rectangular in shape. The largest face, the medium-sized face, and the smallest face of the eraser are rectangular in shape. The smallest face of the pencil is circular, while the largest face is rectangular in shape. Due to the irregular geometry of the pencil sharpener, only the most planar surface was used in the experiment to minimize unpredictable aerodynamic effects.

TABLE III
MASSES OF THE OBJECTS

Objects	m (kg)
Eraser	1.280×10^{-2}
Pencil Sharpener	5.000×10^{-3}
Pencil	1.200×10^{-2}

The masses of the objects were approximated on the basis of literature values due to the unavailability of a precision scale [5]. This approximation may introduce a small systematic error in the calculations. The drag coefficient values used in the air resistance calculation were also obtained from the literature in the same way, since there is not a sufficiently long altitude to achieve terminal velocity.

TABLE IV
THE DRAG COEFFICIENTS OF THE OBJECTS

Objects	C_d
Eraser	≈ 1.17
Pencil	≈ 0.82
Pencil Sharpener	≈ 1.20

C. Experimental Setup

The experimental setup included a ruler, a pencil sharpener, a pencil, an eraser, and a lead pencil. The objects were released from predetermined heights using a stopwatch opened on a laptop computer, and the stopwatch was controlled manually. Due to the limitations of human reflexes, a small margin of error has been factored into the tests². Heights were marked on a vertical surface with a pencil and used. Since the margin of error here is negligible, no height-related margin of error has been included in the calculation.

D. The Tests of Eraser (ET01, ET02, ET03, ET04, ET05, ET06)

The values obtained from Table I, Table II, Table III and Table IV were substituted into equation (11), and 6 different values of t were measured to obtain the g values. To reduce the error margin in each test, four experiments were conducted for each test and their average is reported with an uncertainty of $t \pm 0.010$ s.

1) *ET01*: The smallest surface area of the eraser and was considered the surface area exposed to air resistance, the height is 0.3 m and its fall under the same conditions was measured. $t = 0.259$ s value was obtained. Based on this value, the g value $g_{ET01} = -8.887^{+0.727}_{-0.651}$ m/s², has been found.

2) *ET02*: The medium-sized surface area of the eraser was considered the surface area exposed to air resistance, the height is 0.3 m and its fall under the same conditions was measured. $t = 0.219$ s value was obtained. Based on this value, the g value, $g_{ET02} = -12.348^{+1.200}_{-1.361}$ m/s², has been found.

3) *ET03*: The largest surface area of the eraser was considered the surface area exposed to air resistance, the height is 0.3 m and its fall under the same conditions was measured. $t = 0.222$ s value was obtained. Based on this value, the g value $g_{ET03} = -11.860^{+0.995}_{-1.151}$ m/s², has been found.

4) *ET04*: The smallest surface area of the eraser was considered the surface area exposed to air resistance, the height is 1 m and its fall under the same conditions was measured. $t = 0.483$ s value was obtained. Based on this value, the g value $g_{ET04} = -8.389^{+0.336}_{-0.357}$ m/s², has been

found.

5) *ET05*: The medium-sized surface area of the eraser was considered the surface area exposed to air resistance, the height is 1 m and its fall under the same conditions was measured. $t = 0.431$ s value was obtained. Based on this value, the g value, $g_{ET05} = -10.301^{+0.469}_{-0.486}$ m/s², has been found.

6) *ET06*: The Largest surface area of the eraser was considered the surface area exposed to air resistance, the height is 1 m and its fall under the same conditions was measured. $t = 0.419$ s value was obtained. Based on this value, the g value, $g_{ET06} = -10.406^{+0.476}_{-0.515}$ m/s², has been found.

E. The Tests of Pencil Sharpener (PST01, PST02)

The values obtained from Tables I, Table II, Table III, and Table IV were inserted into equation (11) to find g , and the t value was measured at 2 different positions(*PST01*, *PST02*). The single face of the sharpener that was considered to have a regular geometric shape was used as the surface area A and was measured with an uncertainty of $t \pm 0.010$ s.

1) *PST01*: *PST01* was measured at a height of 0.3 m and under the same conditions as the other experiments, and t was measured to be 0.252 s and g was found $g_{PST01} = -9.327^{+0.843}_{-0.919}$ m/s²

2) *PST02*: *PST02* was measured at a height of 1 m and under the same conditions as the other experiments, and t was measured to be 0.420 s and g was found $g_{PST02} = -10.856^{+0.519}_{-0.549}$ m/s²

F. The Tests of Pencil (PT01, PT02, PT03, PT04)

The values obtained from Table I, Table II, Table III and Table IV were substituted into equation (11), and 4 different values of t were measured to obtain the g values. To reduce the error margin in each test, all tests were measured with an uncertainty of $t \pm 0.010$ s.

1) *PT01*: *PT01* was conducted by releasing the largest surface of the pencil from a height of 0.3 m. The t value was measured at $t = 0.213$ s and the g value has been found at $g_{PT01} = -12.590^{+1.094}_{-1.269}$ m/s²

2) *PT02*: *PT02* was conducted by releasing the smallest surface of the pencil from a height of 0.3 m. The t value was measured at $t = 0.194$ s and the g value has been found at $g_{PT02} = -15.904^{+1.508}_{-1.784}$ m/s².

²The margin of error was determined as $t \pm 0.010$ s in the tests.

3) *PT03*: *PT03* was conducted by releasing the largest surface of the pencil from a height of 1 m. The t value was measured at $t = 0.442$ s and the g value has been found at $g_{PT03} = -8.602^{+0.371}_{-0.406}$ m/s².

4) *PT04*: *PT04* was conducted by releasing the smallest surface of the pencil from a height of 1 m. The t value was measured at $t = 0.455$ s and the g value has been found at $g_{PT04} = -9.593^{+0.413}_{-0.447}$ m/s².

g_K :Kinematic equations' value
 g_G :Universal Law of Gravity's value

The effect of air resistance on the resulting value as a percentage has also been calculated:

TABLE VI
EFFECT OF AIR RESISTANCE ON EXPERIMENTAL RESULTS

Experiment	a_{air} (m/s ²)	g_{exp} (m/s ²)	Drag Effect (%)
ET01	0.058	-8.887	0.65
ET02	0.162	-12.348	1.31
ET03	0.314	-11.860	2.65
ET04	0.184	-8.389	2.19
ET05	0.466	-10.301	4.52
ET06	0.986	-10.406	9.48
PST01	0.121	-9.327	1.30
PST02	0.482	-10.856	4.44
PT01	0.634	-12.590	5.03
PT02	0.038	-15.904	0.24
PT03	1.635	-8.602	19.01
PT04	0.068	-9.593	0.71
Average Drag Effect			4.29%

As seen in Table VI, the effect of air resistance was determined to increase or decrease in direct proportion to the surface areas of the objects that come into contact with the air during their fall. This feature particularly stands out in the *PT03* and *ET06*. However, considering the proximity of the obtained values to the theoretical value despite these resistance levels, it can be said that air resistance had an effect that was negligible for the objects used in this experiment.

V. DISCUSSION

In the conducted tests, the consistency of the Universal Law of Gravitation with kinematic equations was determined to be a satisfactory result. As seen in the other manuscripts in literature, there is no article that explain this consistency with including air resistance effect. In this article, the differential equations of air resistance were included to gain g value more sensitive. Considering that the tests were performed by human hands and with amateur equipment, the result is quite consistent. The consistency of the kinematic equations with gravitation represents the compatibility between the most fundamental laws of classical mechanics. On the other hand, the effect of air resistance on these tests is negligible compared to the surface areas of the experimental equipment. In fact, it is not possible to ignore this effect for objects where the width is much greater than the length. Examples include bird feathers, papers, or parachutes. When approached from a theoretical perspective, the variability of the controlled variables in the experiment has a negligible effect on the experimental result.

VI. CONCLUSION

Based on the experimental results, The Universal Law of Gravitation is largely consistent with the Kinematic Equations (Consistency is 83.20%). Of course, it is found with the error

IV. RESULTS

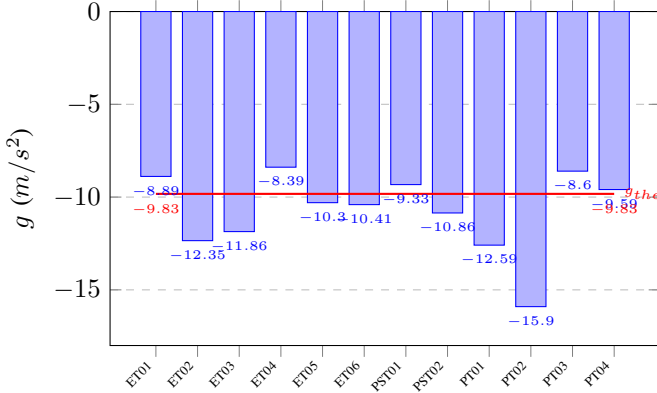


Fig. 1. Measured gravitational acceleration values for all experiments. The red line indicates the theoretical value $g \approx -9.827$ m/s².

When the experiments were compared, it was determined that small surfaces and heights of 1 m were much more consistent with the value obtained from the Law of Gravitation(1-2). If the acceleration value in the law of gravitation is considered the theoretical value and the acceleration value in kinematic tests is considered the actual value;

TABLE V
PERCENT ERROR ANALYSIS OF ALL EXPERIMENTS

Experiment	g_{exp} (m/s ²)	Error (%)
ET01	-8.887	9.57
ET02	-12.348	25.66
ET03	-11.860	20.69
ET04	-8.389	14.63
ET05	-10.301	4.83
ET06	-10.406	5.90
PST01	-9.327	5.10
PST02	-10.856	10.48
PT01	-12.590	28.12
PT02	-15.904	61.84
PT03	-8.602	12.47
PT04	-9.593	2.38
Average Error		16.80%

As seen in Table V, the consistency of the conducted tests with the Universal Law of Gravitation was calculated in (12).

$$\text{Consistency} : \left(1 - \frac{|g_K - g_G|}{g_K}\right) \times 100 = \%83.20 \quad (12)$$

margin of human hand reflexes. Theoretically and dimensionally, The Universal Law of Gravitation is 100% consistent with the kinematic equations. So it shows that the experimental results can be very different due to measuring type. Lastly, in Table V, the g value of ET02, PT01 and PT01 were far more than theoretical value. It is because the huge effect of margin error of human reflexes.

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